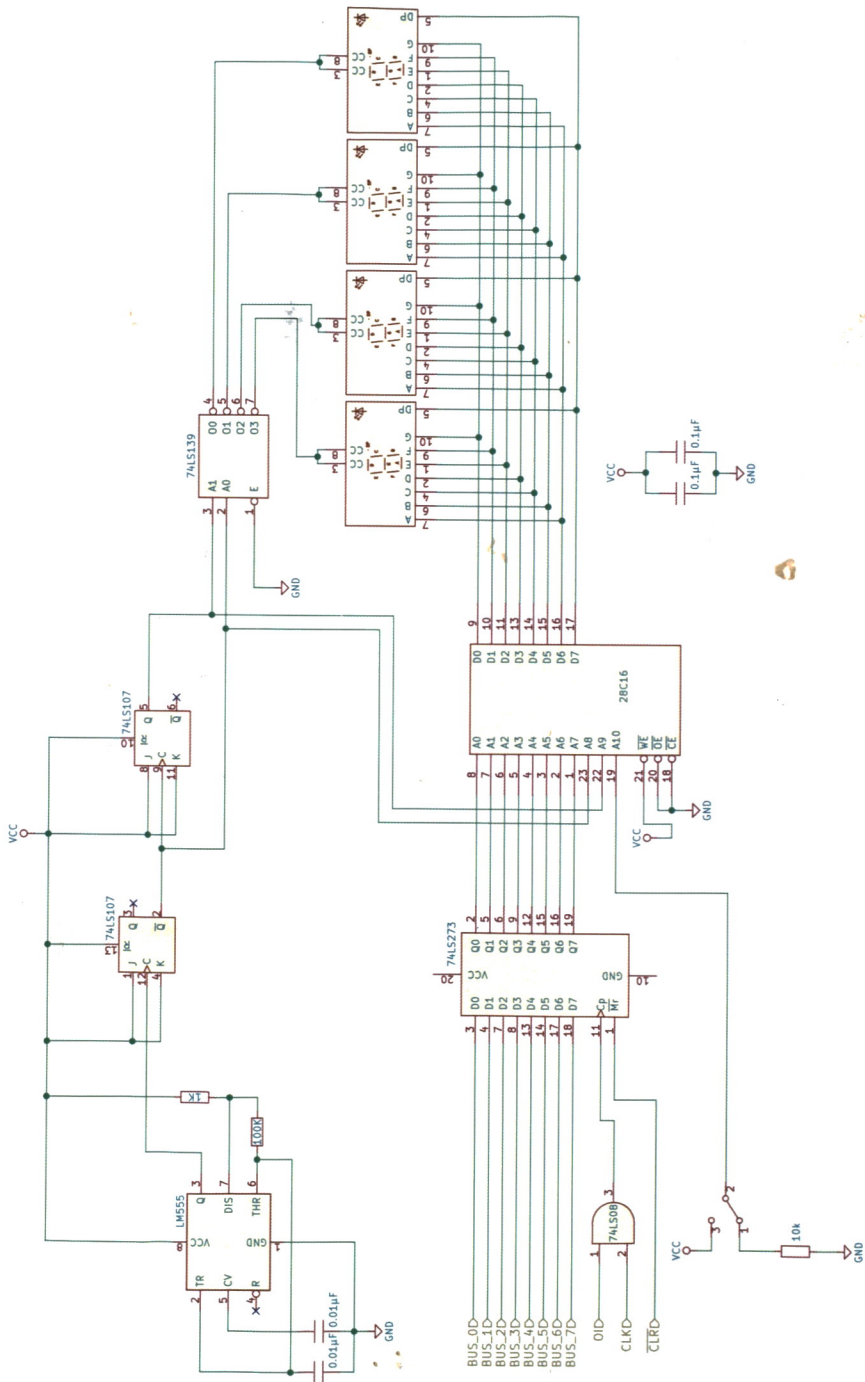
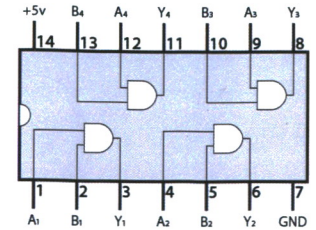


Output module schematic



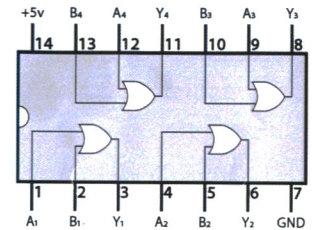
74LS08: AND gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



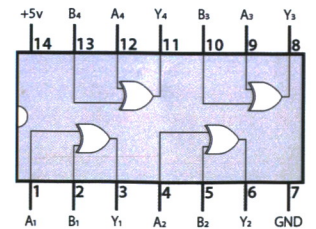
74LS32: OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



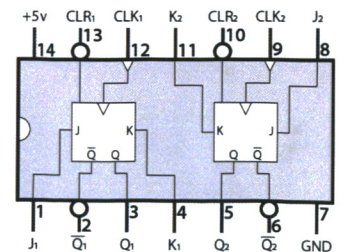
74LS86: XOR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



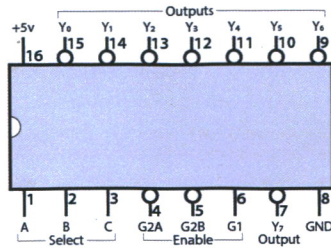
74LS107: JK flip-flop

This chip contains two JK flip-flops. *Note that this is a substitute for the obsolete 74LS76 used in the video.* **Be careful: The pinout is different!**



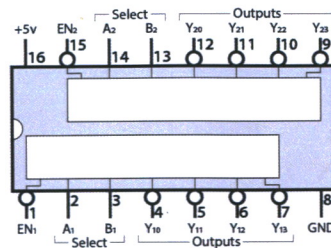
74LS138: 3-to-8 line decoder/demux

This decodes the 3-bit A/B/C select inputs and enables (active low) the corresponding one of the 8 output lines.



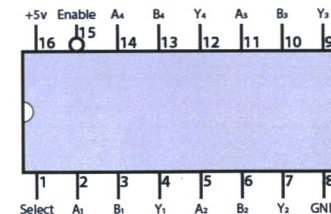
74LS139: 2-to-4 line decoder/demux

Each of the two decoders decodes two "select" input bits to enable the corresponding one of the 4 output lines.



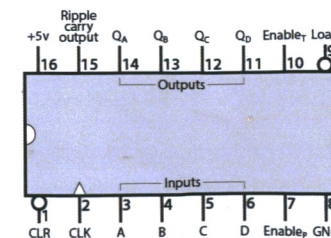
74LS157: 2-to-1 line selector

A 4-bit input is selected from one of two sources (A or B) and is routed to the four outputs (Y).



74LS161: 4-bit binary counter

The 4-bit output is incremented on each CLK pulse. Alternately, any value can be loaded via the 4-bit input.



1 Momentary pushbutton



1 Slide switch



This replaces the pushbutton toggle switch used in the videos.

40 220 Ω resistors (red-red-brown)



Includes 6 extra

5 1k Ω resistors (brown-black-red)



Includes 3 extra

10 10k Ω resistors (brown-black-orange)



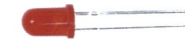
Includes 1 extra

5 100k Ω resistors (brown-black-yellow)



Includes 4 extra

20 Red LEDs



Includes 9 extra

10 Green LEDs



Includes 3 extra

20 Blue LEDs

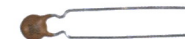


Includes 4 extra

4 7-segment LED display



5 0.01 μ F capacitors (103)



Includes 3 extra

5 0.1 μ F capacitors (104)



Includes 1 extra